Experiment 3

CODE:-

```
#include <stdio.h>

int main() {
   int a, b, c, d, e, f, g, h;
   int m1, m2, m3, m4, m5, m6, m7;
   int C11, C12, C21, C22;

   printf("Enter first 2x2 matrix:\n");
   scanf("%d%d%d%d", &a, &b, &c, &d);

   printf("Enter second 2x2 matrix:\n");
   scanf("%d%d%d%d", &e, &f, &g, &h);

   m1 = (a + d) * (e + h);
   m2 = (c + d) * e;
   m3 = a * (f - h);
   m4 = d * (g - e);
   m5 = (a + b) * h;
   m6 = (c - a) * (e + f);
   m7 = (b - d) * (g + h);

   C11 = m1 + m4 - m5 + m7;
   C12 = m3 + m5;
   C21 = m2 + m4;
   C22 = m1 - m2 + m3 + m6;
```

```
    printf("\nResult Matrix:\n");
    printf("%d %d\n", C11, C12);
    printf("%d %d\n", C21, C22);

    return 0;
}
```

Example

If:

```
A = 1  2
    3  4

B = 5  6
    7  8
```

Output:

```
Result Matrix:
19 22
43 50
```